

5. When  $\text{CCl}_4$  is added to water two separate layers are formed and there is no reaction, whilst  $\text{SiCl}_4$  reacts violently with water.

Give the reason for this difference.

[1]

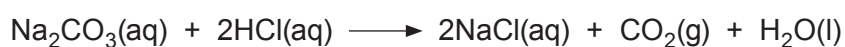
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6. The entropy change of the reaction shown below is positive. Give a reason for this.

[1]



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7. Carbon monoxide can be used to extract iron by the reduction of  $\text{Fe}_3\text{O}_4$ .

(a) Write an equation for this reaction.

[1]

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(b) Give a reason why carbon monoxide acts as a reducing agent while the equivalent oxide of lead does not.

[1]

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8. Draw the **structure** of the copper-containing ion formed when concentrated hydrochloric acid is added to a solution containing  $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$  ions.

[1]

